

Phase 7
Project Documentation Phase

Phase 7: Project Documentation Phase

Phase	Covers
Brainstorming & Ideation Phase	Study management problem & proposed AI StudyBuddy solution
Requirement Analysis Phase	Software & hardware requirements
Project Design Phase	Backend layered architecture & AI service integration
Project Planning Phase	ER diagram, user flow, roles & MVC pattern
Project Development Phase	Backend setup, MongoDB connection, schemas & core code
Project Testing Phase	API endpoint testing & validation
Project Documentation Phase	Code structure, schemas & endpoints compilation
Project Demonstration Phase	Working video demo & live API execution

Project Core Highlights

- **Problem & Requirements** — Explains the study management and content processing challenges faced by students, the proposed **AI StudyBuddy** solution, role-based usage (Student, Teacher/Mentor, Admin), and the required software and hardware setup.
- **Design & Architecture** — Describes the layered backend architecture, Entity-Relationship (ER) database structure, MVC pattern (Model-View-Controller), user access roles, and system workflows.
- **Development & Implementation** — Covers VS Code project setup, Node.js/Express server configuration, MongoDB/Mongoose connection and schemas (User, Note, Quiz), JWT-based authentication, routing, and Google Gemini AI API integration.
- **Testing & Demonstration** — Features end-to-end API testing using Postman/Thunder Client for authentication, study note management, AI text summarization, and automated quiz/flashcard generation, along with a live execution setup.